



Ecole Centrale Nantes, France
Energetics and Propulsion
Masters Program 2022-24

Masters Thesis

Topic: Characterization and modeling of vibration reduction
devices associated with portable machines

August 22, 2024

Presented by: Sahana MARUTHI
Academic Tutor: Dr. Xavier TAUZIA
Institution: ECN
Internship Supervisor: Quentin PIERRON
Organization: INRS



Institut National de Recherche et de Sécurité
Department of Vibrations
France 54519

Abstract

The modern workplace demands attention to occupational safety, especially concerning handheld machinery's vibration-related hazards. This internship explores the characterization and modeling of vibration reduction devices Lateral Flexible Handles [2] associated with portable grinders, addressing the pressing need to control workers exposure to harmful vibrations. Through empirical studies and computational modeling, this internship aims to quantify the effectiveness of various LFH and propose enhancements for optimal performance.

The internship begins with experimental setups and a systematic plan to evaluate commercially available LFH. Measurements using accelerometers positioned at key points on the grinder body and lateral handles provide insights into vibration transmission to operators hands.

Subsequently, to permit extrapolation of the experimental results, to avoid numerous tests with human subjects and to help to improve LFH, dynamic response of LFH are modelled. Furthermore, model parameters are identified from lateral handle stiffness at three specific points determined by static mechanical tests on LFH and from measured weights. Some vibration level under the hand during experimental test are calculated and compared to measured one.

Résumé

Un lieu de travail moderne exige une attention particulière à la sécurité au travail, en particulier en ce qui concerne les risques liés aux vibrations des machines portatives. Ce stage vise à caractériser et la modéliser la réduction des vibrations associée à l'utilisation de poignées latérales flexibles (LFH) pour maintenir des meuleuses portatives. Cela répond au besoin urgent de contrôler l'exposition des travailleurs aux vibrations nocives. Grâce à des études empiriques et à une modélisation numérique, ce stage quantifie l'efficacité de plusieurs LFH et la modélisation permettra de proposer des améliorations pour des performances optimales.

Le stage commence par l'élaboration d'un plan d'expérience pour évaluer des LFH disponibles dans le commerce. Les mesures sont faites à l'aide d'accéléromètres positionnés à des points clés du corps et des poignées de la meuleuse. Elles permettent d'évaluer la transmission des vibrations aux mains des opérateurs.

Par la suite, pour permettre l'extrapolation des résultats expérimentaux, pour éviter de nombreux tests avec des sujets humains et pour aider à la conception de LFH, la réponse dynamique de LFH est modélisée. Des paramètres du modèle sont identifiés à partir de mesures de la raideur statique des LFH en trois points spécifiques et à partir de poids mesurés. Certains niveaux de vibrations sous la main pendant le test expérimental sont calculés et comparés à ceux mesurés.

Thank you

Thank you, Dr. Xavier TAUZIA, for your invaluable guidance and support throughout my Masters Thesis. I would also like to extend my appreciation to Professor Panagiotis Katronis for extensive knowledge in the vibration field. I would extend my thanks to my Internship Supervisor, Mr. Quentin PIERRON, for his continuous support and for making the time to explain everything. Lastly, I would like to thank the entire EVO team at INRS for their collaborative efforts and for creating a pleasant working environment. It has been a privilege to work with such a knowledgeable and supportive group of professionals.

Sincerely,
Sahana MARUTHI
Energetics and Propulsion
Ecole Centrale de Nantes

Contents

Abstract	iii
Résumé	iv
Thank you	v
1 Internship Environment	1
2 Introduction	3
2.1 Motivation	3
2.2 Objective and Problem description	3
2.3 Background	4
2.4 Standard for HAV	5
2.4.1 ISO 5349-1	5
2.4.2 Methods of Vibration Measurement	6
3 Experiment	10
3.1 Literature	10
3.2 Experiment	11
3.2.1 Objectives	11
3.2.2 Experimental setup	11
3.2.3 Grinding Machines	12
3.2.4 Vibration Measurement System	13
3.2.5 Lateral Flexible handles	14
3.2.6 Procedure	16
3.3 Results	18
3.3.1 Grinder body	23
3.3.2 Lateral Handles	25
3.3.3 Handle acceleration under the hand	28
3.3.4 Handle acceleration at the Reference distance	30

3.4	Discussions	33
4	Model	35
4.1	Support hand-arm system	35
4.1.1	Standard ISO 10068	35
4.1.2	Spring-Mass-Damper System Model	36
4.1.3	Analytical formulation	38
4.1.4	Validation of the calculation of the dynamic response	40
4.2	Model of supported Lateral Flexible Handle	44
4.2.1	General description	44
4.2.2	Usage of LS-DYNA finite-element software	47
4.2.3	Usage of Python	47
4.3	Discussions	48
5	Conclusion and Future Works	49

List of Figures

1	INRS research centre in Nancy	2
2	X-axis measurement	6
3	Y-axis measurement	7
4	Z-axis measurement	7
5	Tri-axial Accelerometer	8
6	Frequency-Weighting factor curve	9
7	Experiment setup	11
8	Metabo grinder with all 5 attached accelerometers	12
9	Accelerometer attached to the left side of the LFH	13
10	Flexible Lateral handles	15
11	Three Operators with Standard operating position	16
12	stable vibration levels for all the trials (a)	18
13	stable vibration levels for all the trials (b)	19
14	Operator Impact on vibration levels (a)	20
15	Operator Impact on vibration levels (b)	21
16	Disc effect on vibration level for Bosch grinder	22
17	Disc effect on vibration level for Metabo grinder	23
18	Effect of handles on vibration levels of Bosch grinder	24
19	Effect of handles on vibration levels of Metabo grinder	24
20	Effect of handles on vibration levels of Lateral handle for Bosch grinder	26
21	Effect of handles on vibration levels of Lateral handle for Metabo grinder	27
22	Handles vibration levels for Bosch grinder at the Centre distance	28
23	Handles vibration levels for Metabo grinder at the Centre distance	29
24	Handles vibration levels for Bosch grinder at the Reference distance	31
25	Handles vibration levels for Metabo grinder at the Reference distance	32
26	Standardized values for Model-B	37
27	Apparent mass according x-axis.	41
28	Apparent mass according y-axis.	42
29	Apparent mass according z-axis.	43

Acronyms

FEA Finite Element Analysis. 44, 45, 47, 49

FRF Frequency Response Function. 47, 48

HAV Hand-arm Vibration. 3, 5, 10, 47–50

HAVs Hand-arm Vibration Syndrome. 3–5, 46, 48, 49

HTV Hand transmitted Vibration. 4

INRS National Institution for Research and Safety. i, 1

LFH Lateral Flexible Handles. iii, iv, 3–5, 14, 15, 17, 25, 33, 48–50

OSH occupational safety and health. 1

RMS Root-Mean-Square. 9

SMDs Spring-Mass-Damper System. 35, 36, 38, 40, 44, 45, 47, 48

WBV Whole-body Vibration. 4

1 Internship Environment

The National Institution for Research and Safety in figure 1 is a prominent organization in France dedicated to occupational safety and health. Headquartered in Paris and with a research center located in Nancy, INRS is equipped with advanced facilities and tools essential for conducting experiments and research. The institution plays a pivotal role in preventing occupational accidents and diseases through a multifaceted approach that includes research, training, information dissemination, and expert support.

With a team of approximately 580 professionals from various fields including engineers, doctors, ergonomic specialists, legal advisors, and information experts at INRS provides a rich environment for learning and collaboration. The diverse expertise within the institution enables it to address a broad spectrum of OSH challenges and develop innovative solutions.

Interns at INRS have the opportunity to immerse themselves in this dynamic environment, working in a professional laboratory setting where they engage in research activities, data collection, and analysis. The internship program emphasizes practical experience, allowing interns to contribute to significant projects focused on reducing occupational hazards and improving safety standards.

The Department of Work Equipment Engineering, in particular, offers a stimulating internship experience. Interns in this department work on projects aimed at minimizing risks associated with work equipment, contributing to meaningful advancements in occupational safety. This hands-on involvement provides interns with a comprehensive understanding of real-world applications.



Figure 1: INRS research centre in Nancy

2 Introduction

Vibration exposure is an important occupational hazard that affects countless workers globally, particularly those who operate hand-held power tools [4] as part of their daily work. One of the most concerning problems associated with vibration exposure is Hand-arm Vibration Syndrome. HAVs cause a range of symptoms, including tingling and numbness in the fingers, reduced grip strength, and impaired circulation. Over time, these vibration-induced effects can result in permanent damage to the nerves, blood vessels, and musculoskeletal structures in the hands and arms, significantly impairing a worker's ability to perform manual tasks.

2.1 Motivation

The motivation for this internship stems from the critical need to control the risks associated with HAV exposure. By focusing on the performance of vibration reduction using LFH, this internship aims to evaluate their effectiveness in controlling the risks associated with HAVs. The goal is to quantify the impact of these LFH on vibration levels transmitted to the hands, develop predictive models for vibration transmission, and propose practical interventions to enhance worker safety.

Understanding how different vibration reduction LFH perform in real-world conditions is crucial for improving workplace safety standards. This research will contribute valuable insights into the design and effectiveness of anti-vibration lateral handles, ultimately aiming to reduce the incidence of HAVs and improve the overall health and productivity of workers.

2.2 Objective and Problem description

The primary objective of this thesis is to comprehensively investigate the effects of Hand-arm Vibration on workers using portable power tools. This investigation involves quantifying the magnitude and frequency of vibrations transmitted to the hands and arms during various occupational tasks. By conducting detailed experiments and measurements, the study aims to provide a thorough understanding of how HAV impacts the human hand.

Furthermore, this research seeks the development of predictive models that simulate the transmission of vibration from LFH to the human hand. These models will incorporate

factors such as LFH vibration intensity and transmittance. By refining these models through LS-Dyna and experimental data, the research aims to provide practical LFH for assessing and managing vibration exposure through rigorous testing and analysis, the objective is to propose practical interventions that minimize vibration exposure and mitigate associated health risks effectively.

2.3 Background

Vibration, in the context of occupational health, refers to the mechanical oscillations that are transmitted from machinery and tools to the human body. These oscillations can vary in frequency and amplitude, impacting different parts of the body and leading to various issues. The study of vibration effects has become increasingly important as industries rely more on mechanized tools and equipment. Understanding these effects is crucial for preventing occupational injuries and diseases. Categories of vibration is given below:

- **Hand transmitted Vibration:** Hand-transmitted vibration occurs when vibrations are conveyed from a tool or machine to the hand and arm of a worker [12]. This type of vibration is prevalent in occupations involving the use of tools such as grinders, sanders, and jackhammers. Prolonged exposure to HTV can result in HAVs, a condition characterized by vascular, neurological, and musculoskeletal symptoms.
- **Whole-body Vibration:** Whole-body vibration is transmitted to the body through a supporting surface, such as the seat or platform of a vehicle. WBV is common in occupations involving the operation of heavy machinery, trucks, and construction equipment. The adverse effects of WBV include lower back pain, fatigue, and impaired motor skills, which can significantly affect a worker's health and productivity.

2.4 Standard for HAV

In this internship, the experimental work, measurements and analysis are closely aligned with recognized standards like [3] [6] to ensure that the results are reliable and applicable to real-world scenarios. By following these standards, the internship aims to accurately measure the effectiveness of various anti-vibration LFH's and contribute to the development of safer handles.

One of the key standards used in this research is ISO 5349-1. This standard is instrumental in guiding the vibration testing procedures and ensuring that the evaluation of HAV is conducted in a standardized manner. The following section provides a detailed overview of standard [3] and its relevance to the testing and analysis conducted during the internship.

2.4.1 ISO 5349-1

General requirements, serves as the cornerstone for assessing HAV in occupational settings. This European standard, widely adopted internationally, provides a comprehensive framework for evaluating the impact of vibrations on workers using hand-held power tools, including grinders. The standard's significance in vibration testing of grinder handles stems from its meticulous approach to quantifying vibration exposure.

The primary reasons for employing standard [3] in grinder handle vibration tests are multifaceted. Firstly, it establishes a standardized methodology that ensures consistency and comparability across different studies, equipment, and geographical locations. This uniformity is crucial for researchers and manufacturers alike, as it allows for meaningful bench marking and progress tracking in vibration reduction efforts. Secondly, the standard is specifically tailored for occupational environments, making it particularly relevant for assessing tools like grinders that are predominantly used in professional settings. It takes into account the complex interplay between vibration frequency, amplitude, and duration of exposure, which are all critical factors in understanding the potential health impacts on workers.

Moreover, Standard [3] provides detailed guidelines for assessing health risks associated with HAV exposure. It introduces the concept of daily vibration exposure $A(8)$, which quantifies the average vibration magnitude over a standardized 8-hour working day. This metric is instrumental in evaluating long-term exposure risks and helps in establishing safe usage limits for vibrating tools. By adhering to this standard, researchers and manufacturers can not only assess the current vibration levels of grinder handles but also work towards designing tools that minimize the risk of HAVs and other related occupational health issues.

2.4.2 Methods of Vibration Measurement

The methods for measuring vibrations in grinder handles, as prescribed by the standard [3], are designed to capture a comprehensive and accurate representation of the vibrations experienced by the operator. The measurement process is multifaceted and requires careful attention to several key aspects

1. Directional Measurements: One of the most critical aspects of vibration measurement is the consideration of three orthogonal directions. These directions, denoted as x, y, and z axes, figure numbers 2, 3, and 4 correspond to the anatomical structure of the hand and arm:
 - The x-axis runs parallel to the longitudinal axis of the hand, essentially from the knuckles to the wrist.

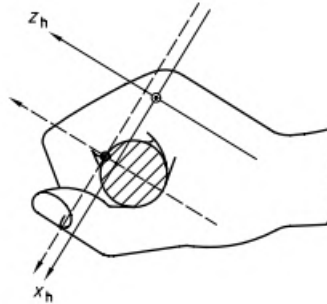


Figure 2: X-axis measurement

- The y-axis is perpendicular to the x-axis, running across the width of the hand.

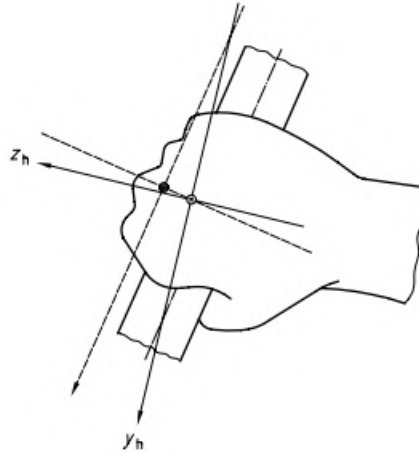


Figure 3: Y-axis measurement

- The z-axis is perpendicular to both x and y axes, running from the palm to the back of the hand.

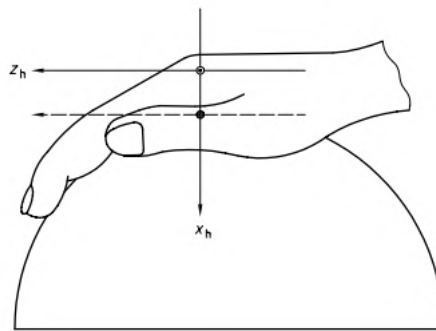


Figure 4: Z-axis measurement

This three-dimensional approach ensures that all components of vibration are captured, as vibrations rarely occur in a single plane.

2. Instrumentation: The primary tool for vibration measurement is the tri-axial accelerometer as mentioned in the below figure number 5. These sophisticated sensors are capable of simultaneously measuring acceleration in all three orthogonal directions. The choice of accelerometer is crucial, as it must have appropriate sensitivity and frequency response characteristics to accurately capture the vibrations typical of grinders. The accelerometer should be as lightweight as possible to minimize any impact on the grinder's dynamics.



Figure 5: Tri-axial Accelerometer

3. Mounting Techniques: The method of attaching the accelerometer to the grinder handle is of paramount importance. The mounting must be rigid to ensure that the vibrations are transmitted faithfully from the handle to the sensor without any damping or resonance effects. Common mounting methods include direct screw mounting, adhesive bonding, or the use of specially designed adapters. The chosen method must not interfere with the normal operation of the grinder or alter the grip of the operator.
4. Measurement Duration and Conditions: Vibration measurements should be conducted over a period that is representative of typical usage patterns. This often involves measuring over complete work cycles or for durations specified in the standard. The grinder should be operated under conditions that reflect real-world usage, including

typical applied forces and working materials. Multiple measurements may be necessary to account for variability in operating conditions.

5. Frequency Weighting: Raw vibration data is processed through frequency weighting filters as specified in the standard [3]. This weighting accounts for the varying sensitivity of the human hand to different vibration frequencies. The standard provides specific weighting curves like in the figure number 6 that emphasize frequencies known to be most harmful to the human hand-arm system.

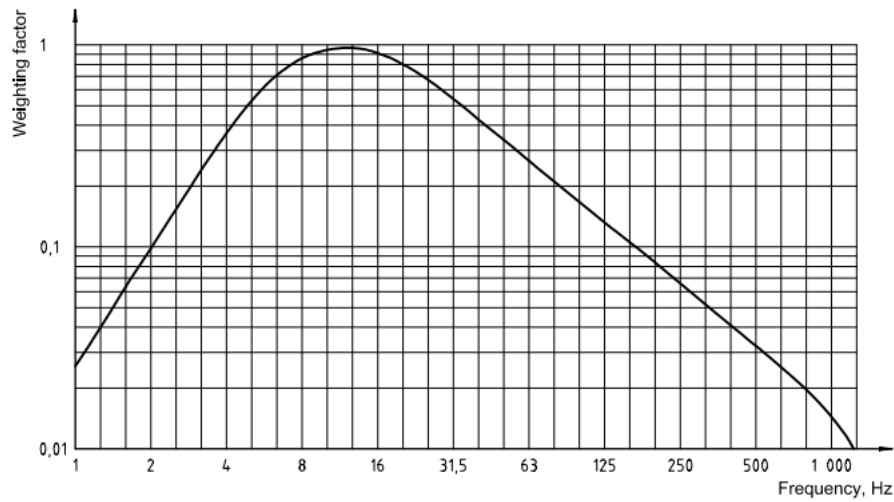


Figure 6: Frequency-Weighting factor curve

6. Data Analysis and Reporting: The collected data is analyzed to determine the frequency-weighted Root-Mean-Square acceleration values for each axis. These are then combined to calculate the vibration total value, which represents the overall vibration magnitude. This value is used to determine the vibration exposure.

3 Experiment

3.1 Literature

Several studies have explored the effects of vibration on the hand-arm system, each employing various methodologies and focusing on different aspects of vibration exposure. This section highlights key research relevant to the study of HAV and identifies connections to and distinctions from the present experiment. For instance, research on the effect of mechanical arm systems on portable grinders utilized accelerometers to measure vibration levels at multiple points, including tool handles and instrumented handles equipped with grip force sensors [11]. This approach is similar to our experiment, which employs a multi-point vibration measurement system to assess vibration transmission through different handle designs. However, while the referenced study focused on mechanical arm support, our experiment specifically evaluates the performance of various handle designs and their effectiveness in reducing vibration exposure for operators.

Another study investigated the acute effects of vibration from chipping hammers and grinders on the hand-arm system, using electrodynamic vibrators to simulate vibration exposure and measure grip and push forces [9] [13]. This aligns with our objective of measuring vibration levels under simulated operational conditions, although our research directly compares different handle designs rather than simulating exposure. Additionally, research on grip pressure distribution under static and dynamic loading highlighted the limitations of various sensors for accurate measurement [7], informing our choice of measurement tools. Our experimental setup, which uses steel unbalanced discs [5] to provide controlled vibrations, allows for consistent input vibrations across different handle configurations, aligning with previous methodologies while focusing on evaluating practical handle designs in real-world conditions [12] [8].

3.2 Experiment

Our experimental approach involves a multi-point vibration measurement system, allowing us to capture a detailed profile of vibration transmission through the grinder and its lateral handle. By comparing these measurements across different lateral handle designs and simulated operational conditions, we aim to identify the most effective strategies for vibration reduction.

3.2.1 Objectives

- Measure the vibration levels of grinding machines under various lateral handle usage.
- Assess the effectiveness of different lateral handle designs in reducing vibration exposure for operators.

3.2.2 Experimental setup



Figure 7: Experiment setup

Our experimental setup in the figure 7 was designed to comprehensively measure and analyze the vibration characteristics of industrial electric grinding machines under various lateral handle configurations. We carefully selected each component of the setup to ensure accurate and reliable data collection. The operator maintained a normal standing posture to replicate realistic working conditions. Vibration levels were recorded for each test condition, with multiple measurements taken to ensure accuracy and consistency. We used Steel unbalanced discs to provide controlled input vibrations, rather than actual grinding discs. These discs allowed us to maintain consistent input vibrations across different lateral handle configurations.

3.2.3 Grinding Machines



Figure 8: Metabo grinder with all 5 attached accelerometers

We employed two models of industrial electric grinders to represent a range of common tools used in professional settings. The first model was the Metabo W12-125 Quick in the figure 8, a compact and versatile grinder commonly used for smaller-scale grinding tasks. This model features an 8mm handle screw diameter, allowing for the attachment of various lateral handle designs. The Metabo W12-125 Quick is known for its high power-to-weight ratio, making it representative of modern, ergonomic grinder designs.

The second model we used was the Bosch GWS 24-230 LVI, a larger and more powerful grinder typically used for heavy-duty grinding operations. This model is equipped with a 14mm handle screw diameter, accommodating larger, more robust lateral handle designs. The Bosch GWS 24-230 LVI represents the class of grinders used in more demanding industrial applications, providing a contrast to the smaller Metabo model in terms of size, power, and potential vibration characteristics.

3.2.4 Vibration Measurement System

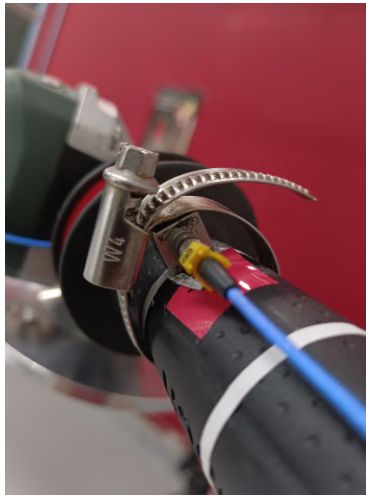


Figure 9: Accelerometer attached to the left side of the LFH

The cornerstone of our vibration measurement system was the PCB 356B20 accelerometer. This high-precision, tri-axial piezoelectric accelerometer is specifically designed for vibration measurements in industrial settings. The PCB 356B20 offers a wide frequency range (2 to 10,000 Hz) and high sensitivity, making it ideal for capturing the complex vibration patterns produced by grinding machines.

We strategically placed five PCB 356B20 accelerometers at 5 key points like in the figure 9 on each grinder to capture a comprehensive vibration profile:

1. Accelerometer L : This sensor was attached to the left side of the lateral handle, oriented to measure vibrations along the Z-axis as mentioned in the figure number 4. This placement allowed us to quantify the vibrations transmitted directly to the operator's hand.
2. Accelerometer R : Positioned on the right side of the lateral handle, this accelerometer also measured along the Z-axis as mentioned in the figure number 4. This placement allowed us to quantify the vibrations transmitted directly to the operator's hand.
3. Accelerometer F : We attached this sensor to the front of the grinder body, measuring along the Y-axis as mentioned in the figure number 3.
4. Accelerometer A : Positioned on top of the grinder body, this accelerometer measured vibrations along the Y-axis as mentioned in the figure number 4.
5. Accelerometer B : Located on the underside of the grinder, this sensor also measured along the Y-axis as mentioned in the figure number 3.

This comprehensive accelerometer arrangement allowed us to capture vibrations in multiple locations, providing a detailed understanding of how vibrations propagate through the tool and into the operator's hands.

3.2.5 Lateral Flexible handles

The primary focus of our study was to evaluate the effectiveness of various LFH designs in reducing vibration transmission to the operator. We tested several anti-vibration handles from leading manufacturers, as well as a standard rigid lateral handle for comparison:



Figure 10: Flexible Lateral handles

1. Metabo Anti-Vibration LFH: Designed specifically for Metabo grinders, this LFH incorporates the manufacturer's proprietary vibration-damping technology.
2. Bosch Anti-Vibration LFH: This LFH features Bosch's vibration control system, designed to minimize vibration transfer in their power tools.
3. Makita Anti-Vibration LFH: Makita offering in vibration-reduction technology, this LFH is designed to be compatible with a range of grinder models.
4. Fein Anti-Vibration LFH: Known for their precision engineering, Fein anti-vibration LFH represents another approach to vibration damping in power tools.
5. Rigid LFH: We included a standard rigid LFH as a control, allowing us to quantify the relative effectiveness of the anti-vibration designs.

Each of these LFH represents a different approach to vibration reduction, incorporating various materials, damping mechanisms, and ergonomic designs. By testing this range of LFH, we aimed to identify the most effective strategies for minimizing vibration transmission to the operator's hands.

3.2.6 Procedure

- Machine setup: Each grinding machine was securely mounted on a stable metal work-bench as in the figure 7. The machines were installed according to manufacturer instructions. For each test run, we installed the selected lateral handle onto the grinder.
- Accelerometer attachment: The five accelerometers were carefully attached to their designated positions on the grinder and lateral handles in the figure 8. We ensured proper orientation of each accelerometer to accurately capture vibrations along the intended axes.
- Operational Positioning: The operator assumed a normal standing work posture as in the figures 11a, 11b, and 11c to simulate realistic working conditions. This standardized posture helped maintain consistency across different test runs and operators.

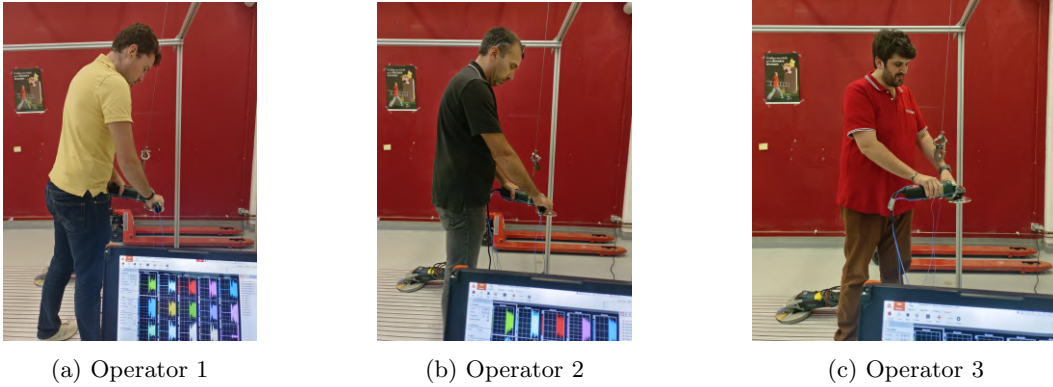


Figure 11: Three Operators with Standard operating position

- Vibration Measurement: With the grinder running at its standard operational speed, we recorded vibration measurements from all five accelerometers simultaneously. Each test condition was measured multiple times to ensure data reliability and to account for any operational variations.
- Testing Conditions: Each machine was tested under different conditions:
 - Various lateral handle designs

- Different grinding machine sizes
 - Different types of grinding discs
 - Different Operators
- Data Analysis: The collected data were analyzed to determine the effectiveness of different lateral handles in reducing vibration and grip force. This helped identify the most effective LFH designs and configurations for minimizing vibration exposure and enhancing operator comfort and safety.

3.3 Results

- Stable performance and Minimal deviation: The low standard deviation across the various accelerometer positions indicates that both grinders exhibited stable performance in terms of vibration levels as we can see in the figures 12 and 13. This means that the vibration intensity did not deviate significantly from trial to trial, suggesting consistent operational behavior.

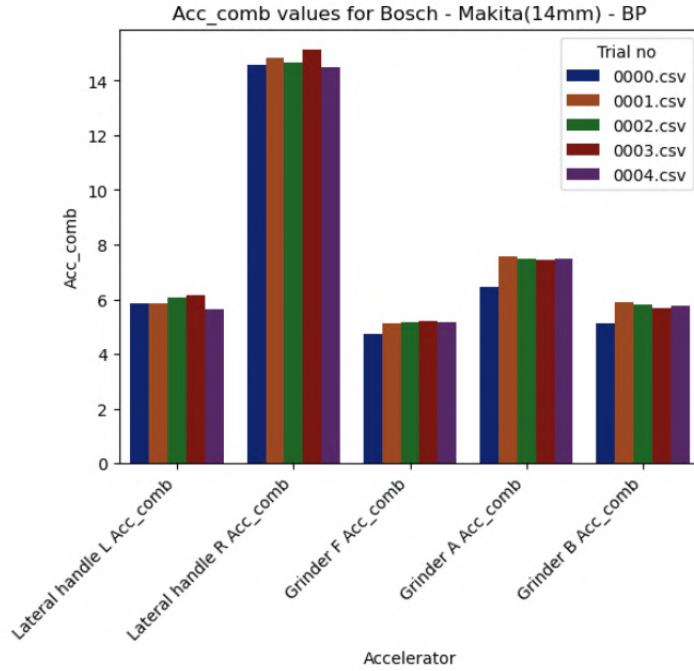


Figure 12: stable vibration levels for all the trials (a)

The figure 12 represents the Bosch grinder with Makita(14mm) handle operated by BP operator with the stable vibration levels for all the trials.

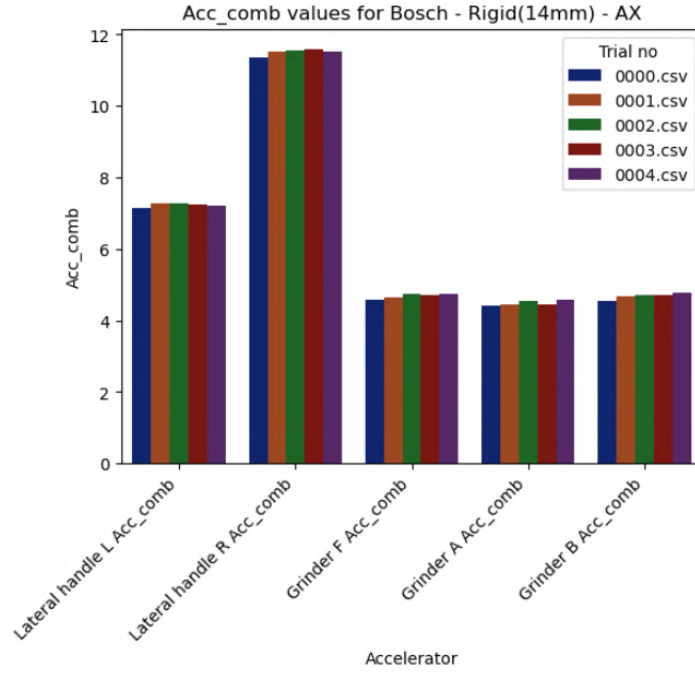


Figure 13: stable vibration levels for all the trials (b)

The figure 13 represents the Bosch grinder with Rigid(14mm) handle operated by Ax operator with the stable vibration levels for all the trials.

- **Operator Impact:** The presence of the operator does not significantly affect the overall vibration levels of the grinding machine as we can see in the figures 14 and 15. However, small differences in vibration values were observed when the operator was using the machines. The slight variations in vibration values can be attributed to the operator's grip and handling style. Although these variations are minimal, they highlight the importance of maintaining a consistent handling technique during the operation to ensure accurate vibration measurements.

The above figure 14 is an example that represents the Bosch grinder with a Makita

(14mm) handle, Operated by AX, BP and Qp operators with the change in the vibration levels due to change in the operator's grip and handling style.

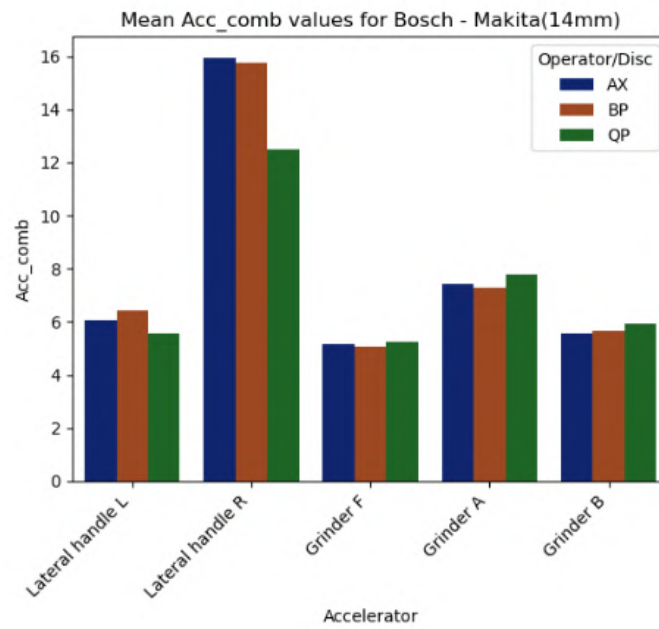


Figure 14: Operator Impact on vibration levels (a)

Similarly, the figure 15 is an example that represents the Bosch grinder with a Metabo (8mm) handle, Operated by AX, MA and Qp operators with the small differences in vibration values due to change in the operator's grip and handling style.

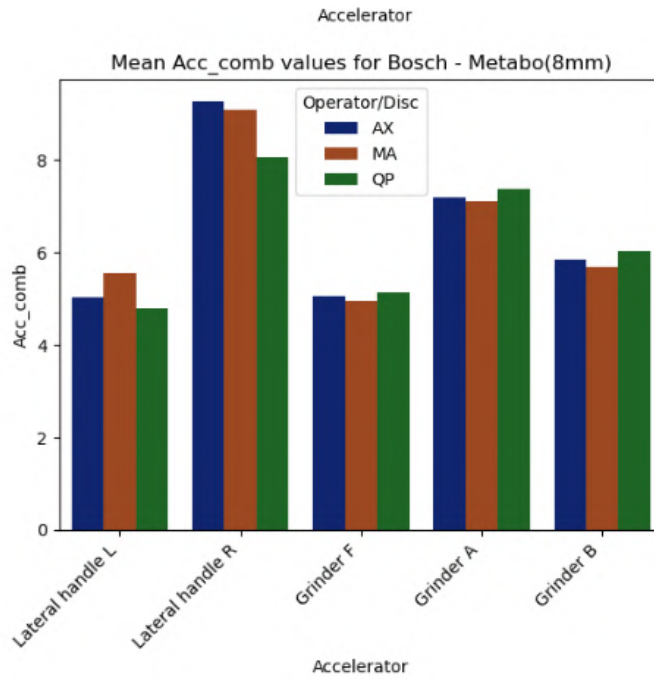


Figure 15: Operator Impact on vibration levels (b)

- Importance of discs: The type of grinding disc plays a crucial role in the vibration behavior of the machines. Specifically, plane and one side raised discs (both balanced and unbalanced) demonstrate significant differences in vibration levels. One side raised discs can create a dynamic balancing effect where the raised portion of the disc may help in redistributing forces more evenly during rotation. This can sometimes lead to reduced vibration levels if the raised section creates a balancing counter-force that mitigates some of the vibrations generated. Balanced discs have a uniform mass distribution, which means that during rotation, the forces are evenly distributed. This uniform distribution reduces oscillatory forces and minimizes vibrations because there are fewer fluctuations in the rotational dynamics. Unbalanced discs have uneven mass distribution, which leads to additional forces during rotation. These additional forces

create oscillations that contribute to higher vibration levels. The imbalance causes the disc to exert varying forces on the grinder, increasing the vibration intensity. Example in the figure 16 is the Bosch Grinder with and Metabo (8mm) handle, Balanced one side raised disc shows very less Vibration levels across all accelerators compared to the Plane disc with both balanced and unbalanced conditions.

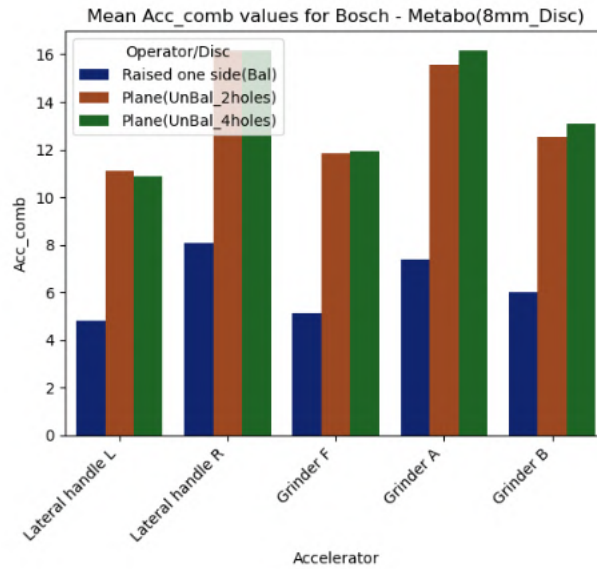


Figure 16: Disc effect on vibration level for Bosch grinder

Example in the figure 17 Metabo grinder with unbalanced plane disc with 2 holes and 4 holes shows significantly higher vibration levels across all accelerators compare to Balanced one side raised disc and This suggests that the Unbalanced Plane disc with holes introduces more imbalance, leading to higher vibration levels compared to balanced plane discs or those with raised sides.

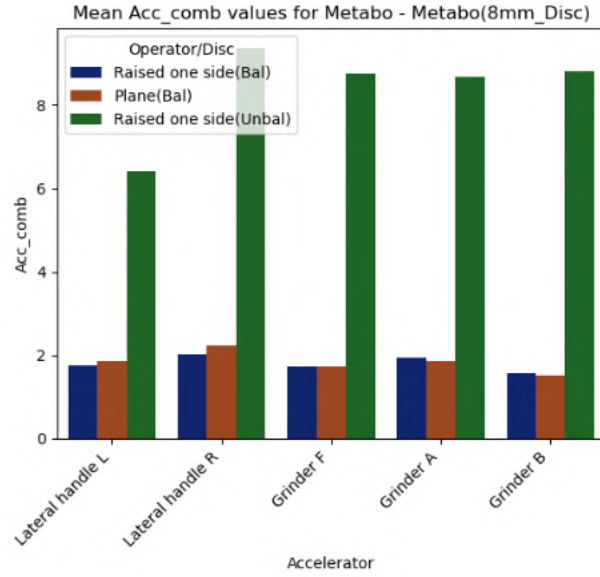


Figure 17: Disc effect on vibration level for Metabo grinder

3.3.1 Grinder body

We examine data from two grinder models Bosch and Metabo. The measurements are taken at three key positions on the grinder body, at F, A, and B positions, which correspond to different areas of the grinder body.

Vibrations originate primarily from the rotating components and the interaction between the grinding wheel and the workpiece. These vibrations propagate through the grinder body and are ultimately transmitted to the operator through the lateral handles and operators hands. The extent and nature of these vibrations can vary depending on several factors, including motor design and power, quality of bearings and other internal components, overall construction and materials used in the body, balance of the grinding wheel speed of operation, and the type of work being performed like grinding, cutting, and polishing etc.

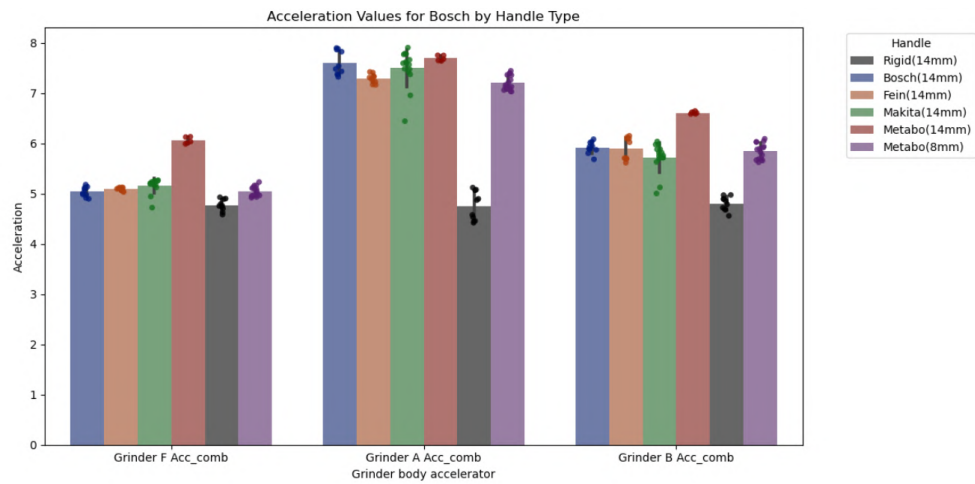


Figure 18: Effect of handles on vibration levels of Bosch grinder

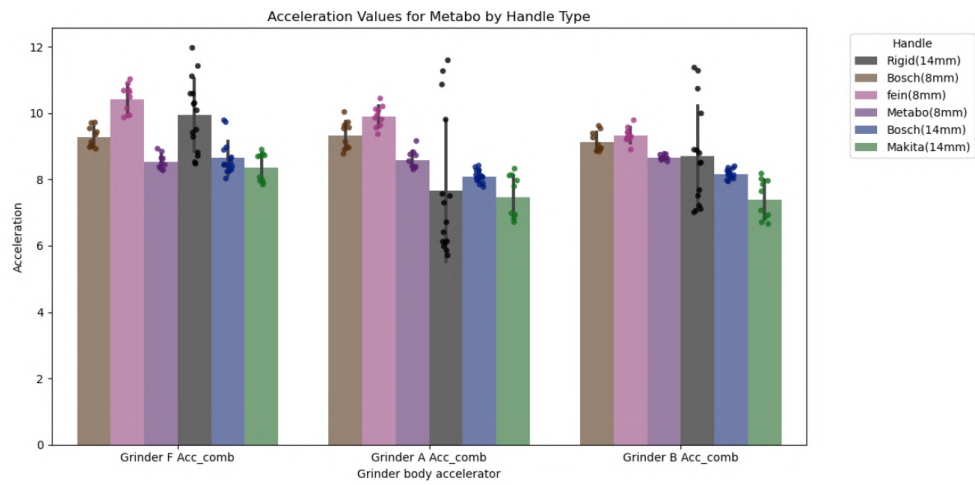


Figure 19: Effect of handles on vibration levels of Metabo grinder

As we can see in the figure 18 Bosch grinder accelerator F values range from 4.68 to 6.06, A values range from 4.49 to 7.70, and B values range from 4.69 to 6.61. For Metabo grinder in the figure 21 accelerator F values range from 7.95 to 10.18, A values range from 6.87 to 9.99, and B values range from 6.84 to 10.12.

Observations:

- Consistency within grinders : For the Bosch grinder, the values are relatively consistent across different lateral handles, with most readings falling within a narrow range. The Metabo grinder shows slightly more variation, but still maintains a level of consistency across lateral handles.
- Position-specific patterns : For the Bosch grinder, the A position generally shows the highest vibration levels, followed by B , then F .For the Metabo grinder, the vibration levels are more evenly distributed across F, A, and B positions.
- Independence from operator and lateral handle : The relative consistency of readings within each grinder model, when considering only the LFH, supports the hypothesis that body vibrations are somewhat independent of operator and lateral handle design. However, the variation we do see suggests that handle design can have a minor influence on grinder body vibrations, particularly for the Metabo grinder. It's important to note that the rigid handle shows a distinct pattern compared to the flexible handles. The rigid handle consistently demonstrates higher vibration levels across all grinder models. This suggests that while body vibrations may be somewhat independent of flexible handle designs, the use of a rigid handle significantly impacts the vibration characteristics of the grinder.

3.3.2 Lateral Handles

In this study, we examine the performance of various lateral handles on two grinder models Bosch and Metabo. The vibration levels are measured at two positions on each handle, labeled as L and R . These measurements help us understand how different handle designs affect vibration transmission and potentially impact user comfort and safety. Lateral handles vary in material (plastic, rubber, or metal), diameter (common sizes include 8mm and 14mm), and length. Some incorporate anti-vibration technologies using springs, dampers, or specialized materials. The choice of handle can significantly influence operators vibration exposure.

Bosch Grinder: As we can see in the figure 20 at L position values range from 4.48 to 8.23, R position values range from 4.51 to 14.73. we can see that The Bosch (14mm) and Fein (14mm) handles perform best for the left and right sides respectively. The Makita (14mm) handle shows a significant difference between left and right sides (6.02 vs 14.73). The Metabo (8mm) handle shows moderate performance on both sides (5.13 left, 8.81 right).

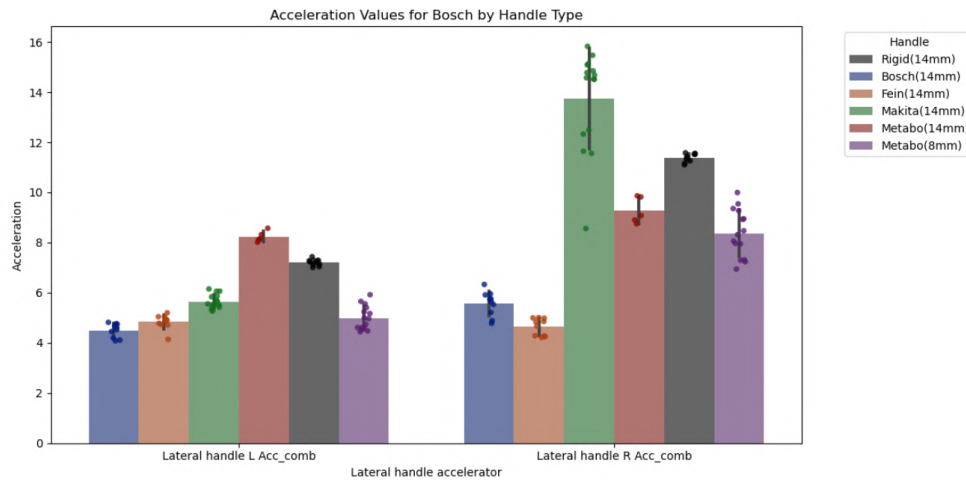


Figure 20: Effect of handles on vibration levels of Lateral handle for Bosch grinder

Metabo Grinder: As we can see in the figure 21 at L position values range from 6.56 to 10.27, R position values range from 3.90 to 24.12. The Metabo (8mm) handle performs best overall for the Metabo grinder. The Makita (14mm) handle shows the best performance on the right side but is among the worst on the left side. The Fein (14mm) handle shows extremely high vibration on the right side (24.12).

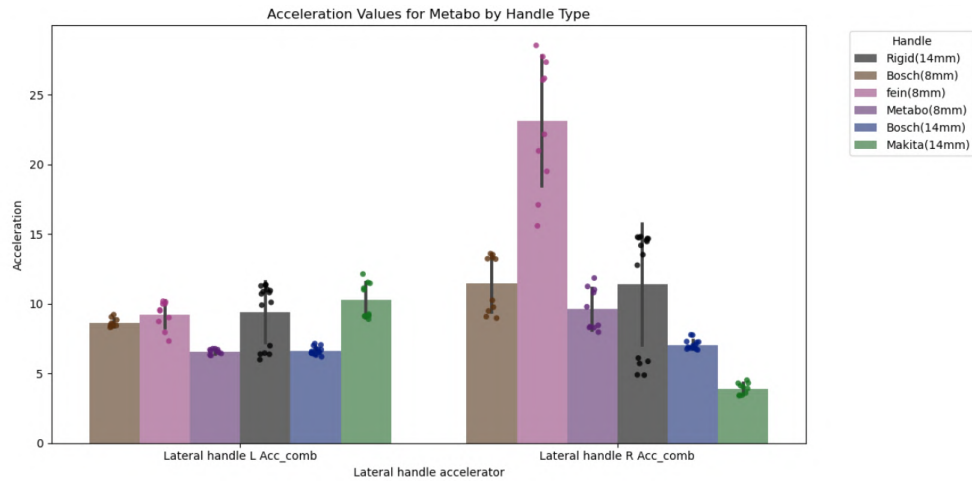


Figure 21: Effect of handles on vibration levels of Lateral handle for Metabo grinder

Observations:

- The 8mm handle (Metabo for Bosch, Metabo for Metabo) generally show more consistent and lower vibration levels across both sides compared to 14mm handles.
- Some handles (Makita 14mm) show large discrepancies between left and right side vibrations, due to large rotation and high distance between the accelerometers
- The rigid handle consistently shows higher vibration levels compared to most flexible handles, supporting the superiority of anti-vibration designs.
- The right handle generally experiences higher vibration levels than the left, especially for the Bosch grinder.
- For the Bosch grinder, the Bosch (14mm) handle for the left side and the Fein (14mm) handle for the right side show the best performance.
- For the Metabo grinder, the Metabo (8mm) handle shows the best overall performance.

3.3.3 Handle acceleration under the hand

Handle acceleration under the hand is defined as the mean accelerations measured by accelerometers L and R. Indeed, during the experimental tests, one hand of the operator is exactly between the accelerometers L and R on the lateral handle. In assuming that lateral handle act as a rigid body between the accelerometers L and R, this acceleration is equal to that which would be measured on the handle under the hand in the middle of the two accelerometers.

Bosch Grinder:

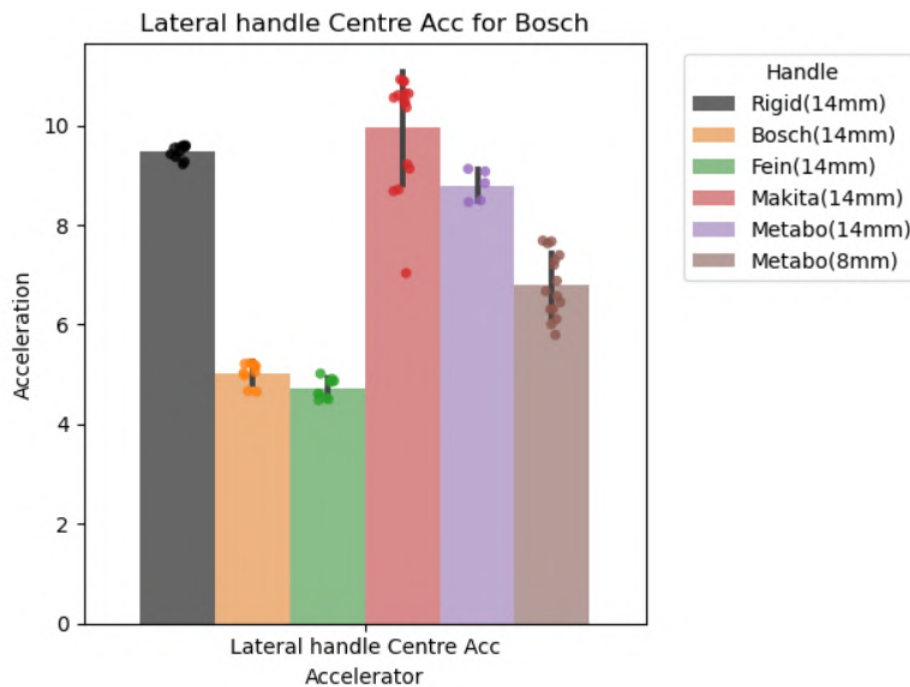


Figure 22: Handles vibration levels for Bosch grinder at the Centre distance

Metabo Grinder:

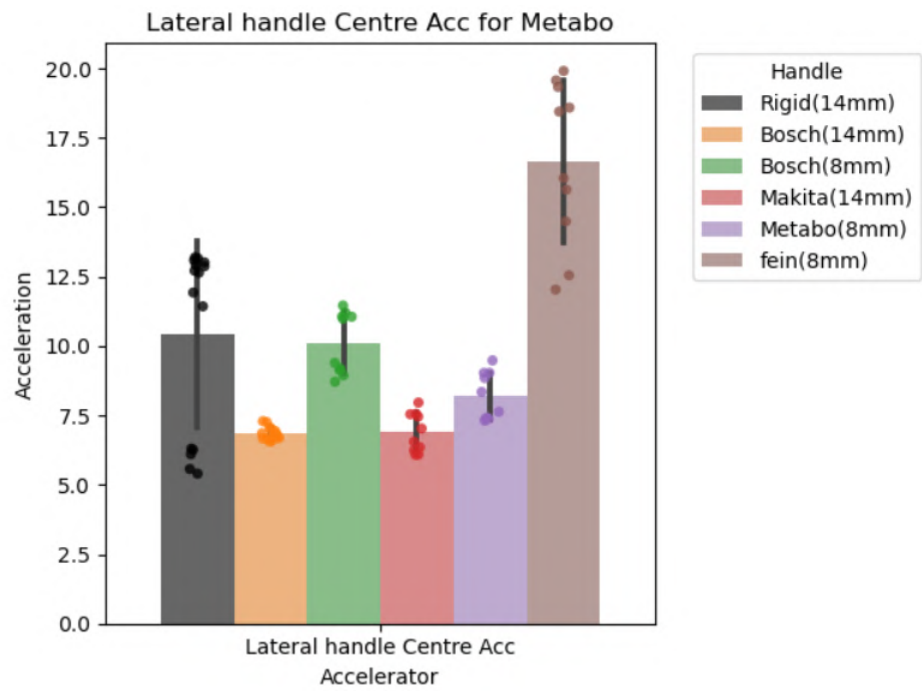


Figure 23: Handles vibration levels for Metabo grinder at the Centre distance

3.3.4 Handle acceleration at the Reference distance

Grinder body rotates and this rotation induce a gradient of acceleration level on the handle according to the distance from the grinder body. To compensate this effect, a unique Reference Distance Mref is defined for all handles. Mref is calculated by determining the average position of the accelerator relative to the starting points of the lateral flexible handles on the grinder. First, measure the distance from the starting point of the left lateral flexible handle, where it is attached to the grinder, to the accelerator. This distance is labeled as D1. Similarly, the distance from the starting point of the right lateral flexible handle to the accelerator is measured and labeled as D2. For each handle, calculate the midpoint by averaging these two distances:

$$\frac{(D1 + D2)}{D2}$$

After calculating the midpoint for all the handles, then compute the average of these midpoint values across all the handles. This average midpoint value serves as the reference distance, Mref. Using this reference distance, by employing the polyfit function in Python to fit a polynomial curve and calculate the acceleration values for the lateral handle at both the left and right positions corresponding to Mref.

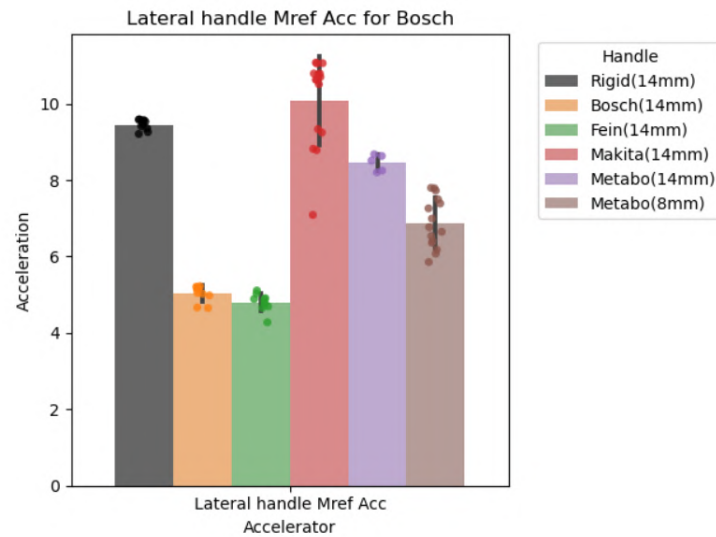


Figure 24: Handles vibration levels for Bosch grinder at the Reference distance

As we can see in the figure 24 Fein and Bosch (14mm) handles perform best for the Bosch grinder. The Metabo (8mm) handle shows consistent performance across different operators. The Makita (14mm) handle shows the highest vibration levels.

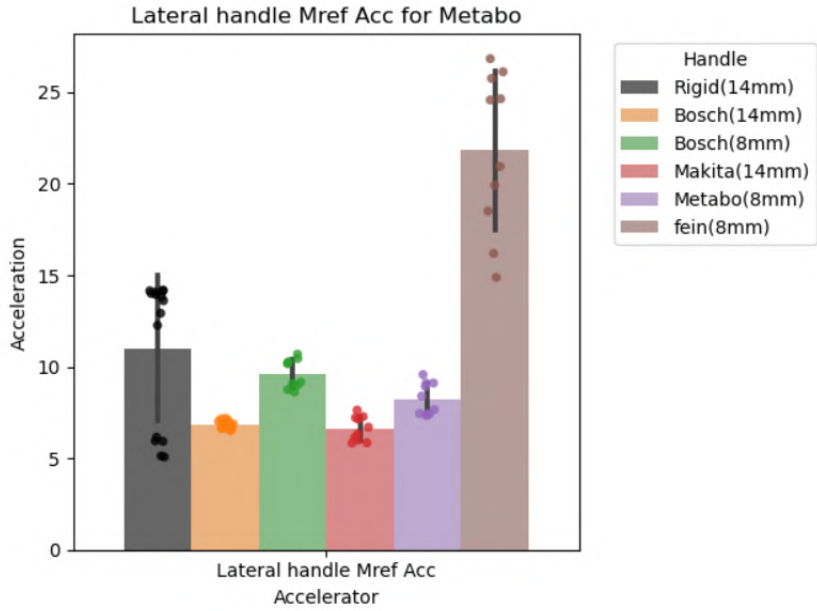


Figure 25: Handles vibration levels for Metabo grinder at the Reference distance

In the figure 24 Makita (14mm), Metabo (8mm), and Bosch (14mm) handles perform best for the Metabo grinder. The Rigid (14mm) handle shows high variability and poor performance. The Fein handle, which performed well on the Bosch grinder, shows higher vibration levels on the Metabo grinder.

The 8mm handles (Metabo for both grinders) generally show consistent performance across different operators. Some handles (Makita 14mm) perform very differently on the two grinders, highlighting the importance of grinder-specific handle design. The rigid handle consistently shows higher vibration levels and more variability across operators/discs compared to most flexible handles.

3.4 Discussions

The tests conducted on various Lateral Flexible Handles for the Bosch and Metabo grinders have yielded important insights into vibration reduction. The primary goal was to determine which LFH most effectively reduce vibration across different grinder models and usage conditions.

For the Bosch grinder, the Fein (14mm) handle reduced vibration by approximately 55% compared to the rigid lateral handle and the Bosch (14mm) handle performed exceptionally well, reducing vibration by about 50% compared to the rigid lateral handle. In the case of the Metabo grinder, the Makita (14mm) handle reduced vibration by approximately 45%, Bosch (14mm) handle reduced vibration by approximately 40%, Metabo (8mm) handle reduced vibration by approximately 30% compared to the rigid lateral handle and the Bosch (8mm) handle performed exceptionally well, reducing vibration by about 15% compared to the rigid lateral handle.

For the grinder body, Bosch grinder with all the LFH shows increased vibration by approximately 10-30% for the grinder front body, 40-45% for above the grinder body, and 20-30% for below the grinder body compared to the rigid lateral handle. In the case of Metabo grinder all the LFH shows increased vibration by approximately 10-20% compared to the rigid lateral handle. and all the LFH shows almost same vibration for respected grinders. from this we can say that the \acrshort{lfh} not impact the vibration levels on the grinder body.

Vibration in the middle of the hand, for Bosch grinder both Fein (14mm) and Bosch (14mm) LFH showed superior performance by reduced vibration approximately by 45-55%, Metabo (8mm) showed reduced vibration approximately by 30% compared to the rigid lateral handle. In the case of Metabo grinder, both Fein (14mm) and Bosch (14mm) LFH showed best overall performance by reduced vibration 40%, Metabo (8mm) showed reduced vibration approximately by 20% compared to the rigid lateral handle. this is due to the effect of difference between LFH level arm and grinder rotation has an influence on vibration transmissibility.

At the reference distance, for Bosch grinder both Fein (14mm) and Bosch (14mm) handles showed superior performance. These LFH reduced vibration by 40-50% compared to the rigid handle. The Metabo (8mm) handle demonstrated consistent and moderate vibration levels. with an average decrease of 30% compared to the rigid lateral handle. In

the case of the Metabo grinder, the Makita (14mm) handle and Bosch (14mm) emerged as the best overall performer for lateral handles by reduced vibration approximately 40-45%. Metabo (8mm) and Bosch (8mm) handles showed good performance by reduction of up to 20-30% in vibration compared to the rigid lateral handle.

An important observation from these tests is the variability in handle performance between different grinders. Handles that performed well on one grinder didn't necessarily show the same level of effectiveness on the other. This highlights the importance of grinder-specific handle design and testing.

Ultimately, these findings emphasize the importance of matching handles to specific grinder models and considering factors such as handle diameter and flexibility in design. Users should consider testing different handles with their specific grinder model and usage conditions for optimal vibration reduction. This research provides valuable guidance for both manufacturers in handle design and for end-users in selecting the most effective handles for their specific needs.

4 Model

In the chapter dedicated to the experiments, vibration level with several lateral handles with two grinders have been measured. In this chapter, we describe, develop and test a mechanical model of lateral handles able to predict the vibration level on their surface during standard experiment. This approach allows us to provide insights for improved tool design. In addition, experimental tests with human subjects could be reduced.

During standard experiment, vibrations of the grinder body are not very affected by operator and lateral handle. Thus, vibrations of the grinder body are considered as an input of models of lateral handles to analyze the impact of different handle designs. The grinders vibration input measured in the previous chapter is used.

Lateral handles are modelled to calculate the vibration under the hand of the operator when the screw of the lateral handle is submitted to the applied vibrations of the grinder body. Model have to allow us to understand how the handles geometry and material properties influence the vibration transmission. LS-DYNA software is used to permit to model complex geometry.

The vibration under the hand depends on the dynamic response of the hand-arm system supporting the grinder by the lateral handle. Thus, support hand-arm system have to be model with lateral handle.

4.1 Support hand-arm system

To simulate the human hand-arm system, we adopt the Spring-Mass-Damper System system as specified in the standard [3]. This standard provides a well-established and validated model for representing the mechanical impedance of the human hand-arm system over a wide frequency range.

4.1.1 Standard ISO 10068

This standard gives mainly mechanical impedance of hand-arm system. It can be used for assessing vibration isolation systems effectiveness and ability to control vibration transmission. By modeling the support arm as a 2 degree of freedom SMDs, we can analyze how

variations in the spring constant, mass, and damping coefficient affects the overall vibration response of the handles. The primary reasons for using the standard [3] in the context of hand-transmitted vibration studies, particularly for tools like grinders, are as follows:

1. **Standardized Approach:** the standard [3] offers a standardized method for representing the mechanical properties of the human hand-arm system. This standardization ensures consistency across different studies and allows for meaningful comparisons between research findings.
2. **Comprehensive Modeling:** The standard provides mechanical impedance data and models that represent the hand-arm system's response to vibration over a wide frequency range (typically 10 to 500 Hz). This range covers most of the significant frequencies encountered in hand-held power tools.
3. **Realistic Representation:** The models in the standard [3] are based on extensive research and empirical data, making them a realistic representation of the average human hand-arm system's response to vibration.
4. **Design and Evaluation Tool:** By using the models provided in the standard [3], engineers and researchers can simulate the interaction between a vibrating tool and the human hand-arm system. This is invaluable for designing tools with reduced vibration transmission and for evaluating the effectiveness of vibration control measures.
5. **Risk Assessment:** The models help in understanding how vibration energy is transmitted to and absorbed by the hand and arm, which is crucial for assessing potential health risks associated with prolonged exposure to hand-transmitted vibration.

4.1.2 Spring-Mass-Damper System Model

the standard [3] presents several Spring-Mass-Damper System models in annex to represent the mechanical impedance of hand-arm system. The model B on the standard was used. This model is particularly useful for understanding and simulating the dynamic behavior of the hand and arm when exposed to vibration. The key aspects of this model are:

1. **Spring-Mass-Damper System :** The hand-arm system is represented as a series of masses connected by springs and dampers. This mechanical analogy allows for a simplified yet effective representation of the complex biological system.

2. **Three-Dimensional Modeling:** The model provides parameters for three orthogonal directions (x, y, and z), corresponding to the anatomical axes of the hand-arm system. This three-dimensional approach ensures that the model captures the directional dependencies of the hand-arm response to vibration.
3. **Multiple Degrees of Freedom:** The model typically includes multiple degrees of freedom, representing different segments of the hand-arm system. This allows for a more nuanced representation of how vibration energy is transmitted through the hand and arm.
4. **Standardized Values:** The coefficients provided in the figure 26 for Model-B are based on averaged data from numerous studies, representing a typical hand-arm system. These standardized values enable researchers and engineers to perform simulations without the need for extensive individual measurements.

Paramètre	Unité	Direction des vibrations		
		x_h	y_h	z_h
m_1	kg	0,5479	0,5374	1,2458
m_2	kg	0,0391	0,0100	0,0742
k_1	N/m	400	400	1000
k_2	N/m	0	17648	50000
c_1	N·s/m	22,5	38,3	108,1
c_2	N·s/m	202,6	75,5	142,4

Figure 26: Standardized values for Model-B

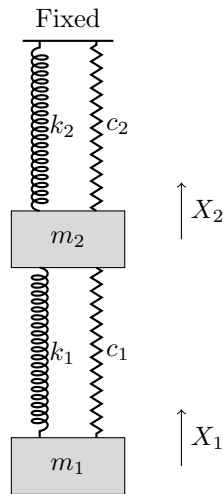
5. **Applicability to Tool Design:** By using these standardized parameters, tool designers can model how their designs will interact with a typical human hand-arm system, allowing for optimization of tool characteristics to minimize vibration transmission.
6. **Impedance and Transmissibility:** The model allows for the calculation of both mechanical impedance and transmissibility. These metrics are crucial for understanding how vibration is transmitted to and through the hand-arm system.

4.1.3 Analytical formulation

We are analyzing Spring-Mass-Damper System to understand the vibration characteristics of a grinder handle. This analysis is crucial because it allows us to model how vibrations are transmitted through the handle to the operator's hand, which is essential for design and vibration control.

1. System Description: The system consists of two masses, springs, and dampers, representing a simplified model of the grinder handle m_1 and the operator's hand-arm system m_2 . This simplification allows us to apply principles of mechanical vibrations to a complex system.
2. Equations of Motion: We derive the equations of motion using Newton's second law for each mass. This step is fundamental because it describes how each part of the system moves in response to forces, including the external force from the grinder.

Analytical solution for the displacements X_2 and X_1 of a SMDs. The system consists of two masses (m_1 and m_2), two springs (k_1 and k_2), and two dampers (c_1 and c_2). Mass m_1 is connected to mass m_2 through k_1 and c_1 , and mass m_2 is connected to a fixed point through k_2 and c_2 . we aim to find the displacements X_1 and X_2 .



For mass m_1 :

$$m_1\ddot{X}_1 + c_1(\dot{X}_1 - \dot{X}_2) + k_1(X_1 - X_2) = f(t) \quad (1)$$

For mass m_2 :

$$m_2\ddot{X}_2 + c_2\dot{X}_2 + k_2X_2 + c_1(\dot{X}_2 - \dot{X}_1) + k_1(X_2 - X_1) = 0 \quad (2)$$

These equations capture how the motion of each mass is influenced by its own properties and its interaction with the other mass.

3. Matrix Representation: We express the system in matrix form because it provides a compact and systematic way to represent and solve coupled differential equations. The matrix form is particularly useful for complex systems and facilitates the application of linear algebra techniques in our analysis.

$$M\ddot{X} + C\dot{X} + KX = F \quad (3)$$

Where:

$$M = \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}, \quad C = \begin{bmatrix} c_1 & -c_1 \\ -c_1 & c_1 + c_2 \end{bmatrix}, \quad K = \begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 + k_2 \end{bmatrix}$$

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \quad F = \begin{bmatrix} f(t) \\ 0 \end{bmatrix}$$

This equation represents the system behavior at any given frequency ω allowing us to analyze how different frequencies of vibration affect the system.

4. Frequency Domain Analysis: We transform the system to the frequency domain for several reasons like it simplifies the differential equations into algebraic equations. It allows us to analyze the system's response to different frequencies of vibration, which is crucial for understanding how the grinder handle performs across its operational range and it facilitates the calculation of important metrics like transmissibility.

The frequency domain equation is:

$$-\omega^2 MX + j\omega CX + KX = F \quad (4)$$

For a harmonic input where X_1 and X_2 known:

$$X_1(t) = X_1 e^{j\omega t}$$

$$X_2(t) = X_2 e^{j\omega t}$$

The system becomes:

$$(-\omega^2 M + j\omega C + K) \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} f(t) \\ 0 \end{bmatrix} \quad (5)$$

This equation represents the system behavior at any given frequency ω , allowing us to analyze how different frequencies of vibration affect the system.

4.1.4 Validation of the calculation of the dynamic response

In this section, we validate the calculation of the dynamic response developed in the previous section. We first derived the equations of motion for the hand-arm system using a two degrees of freedom Spring-Mass-Damper System, as outlined in the previous section. These equations describe how the system responds to vibrations and were formulated in matrix form. Although mass, damping, and stiffness matrices are known from model parameters, we use LS-DYNA to obtain them to be prepared to solve more complex model with the flexible handle. These matrices capture the mechanical properties of the hand-arm system and were essential for simulating its dynamic behavior.

With the matrices obtained from LS-DYNA, we transitioned to the frequency domain analysis using Python. The Python code solves for the system response at various frequencies and computes the apparent mass, a key metric to assess vibration transmissibility. The code processes the system matrices and simulates the dynamic behavior of the hand-arm system across a frequency range from 10 to 500 Hz.

After the analysis, we obtained frequency response curves for the x, y, and z axes of the hand-arm system. These curves were compared to the standard data provided in the Model B of the standard [3]. The results closely match the standardized plots, confirming that our model accurately captures the vibration dynamics of the hand-arm system.

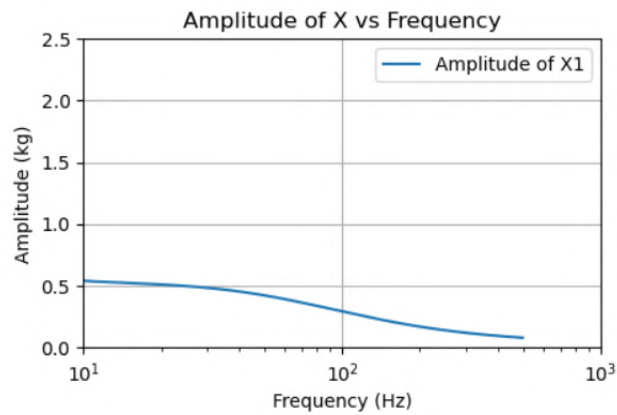


Figure 27: Apparent mass according x-axis.

This figure 27 shows the vibration characteristics in the fore-and-aft direction of the system. As we look at the graph, we see the frequency on the horizontal axis and the amplitude on the vertical axis. The curve generally starts low at very low frequencies, then rises sharply to a peak in the low-frequency range. This peak represents the primary resonance of the system in the x-direction. After this peak, the curve tends to decline as the frequency increases, with possibly some smaller peaks at higher frequencies representing secondary resonances.

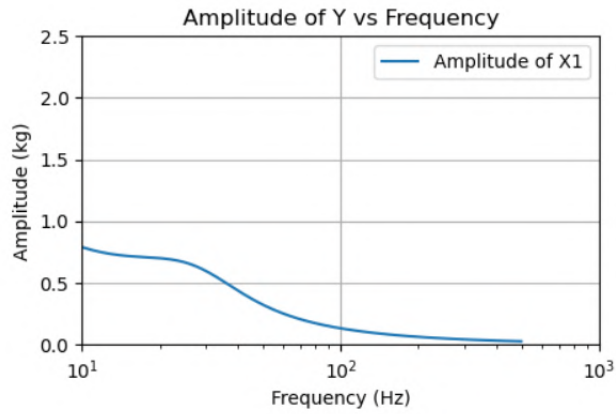


Figure 28: Apparent mass according y-axis.

The shape of this curve 28 is often similar to the x-axis plot, but with some key differences. The initial peak might occur at a slightly different frequency. This lower frequency for the primary resonance in the y-direction is often due to the lower stiffness of system in the lateral direction compared to the fore-and-aft direction. After this main peak, the curve generally declines with increasing frequency, reflecting the complex structural dynamics in the lateral direction.

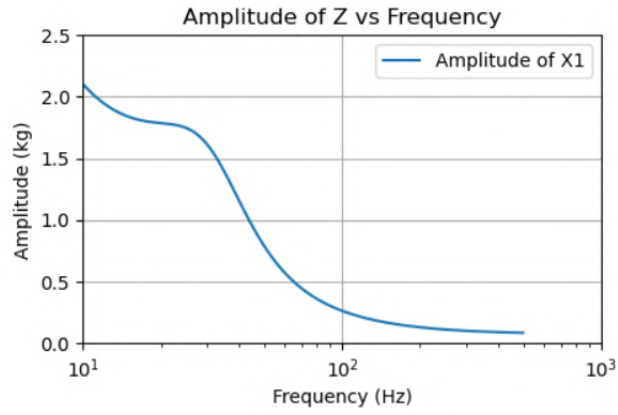


Figure 29: Apparent mass according z-axis.

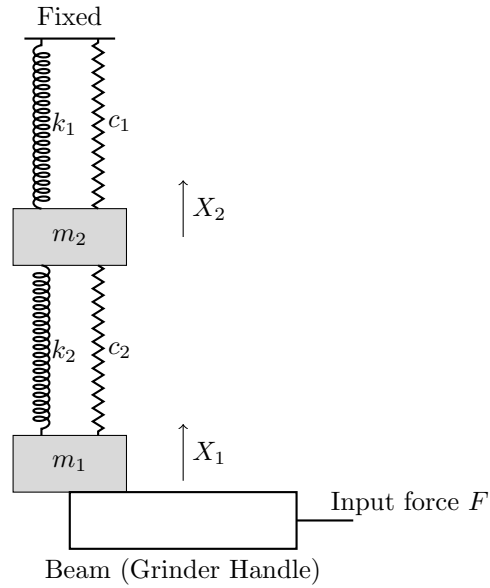
The figure 29 represents vertical vibrations. This plot often shows the most complex behavior of the three. The curve typically starts with a very prominent peak in the low-frequency range. This peak corresponds to the primary vertical resonance of the system. Following this major peak, the curve usually shows a steep decline. These secondary features represent various anti-resonances in the vertical direction. At higher frequencies, the curve generally continues to decline, but may show some additional small peaks corresponding to component resonances.

4.2 Model of supported Lateral Flexible Handle

4.2.1 General description

1. Beam with the Spring-Mass-Damper System (SMDs) : We connect the SMDs to a beam representing the grinder handle. This step is crucial because it models how the vibrations from the grinder are transmitted through the handle (beam) to the operators hand-arm system (SMDs).

To accurately model the beam (grinder handle), we use the FEA [14] method. This approach allows us to discretize the continuous beam into a finite number of elements, each with its own mass, stiffness, and damping properties.



The FEA is particularly useful for complex geometries and non-uniform material properties, which are common in real-world applications like grinder handles. By using this method, we can more accurately represent the beam's behavior under various loading conditions and frequencies.

We split the system into two parts: the beam (represented by finite elements) and the hand-arm system (represented by the SMDs). This splitting allows us to model the interaction between these two subsystems more accurately. This splitting is done to accurately represent the different dynamics of these two subsystems and their interaction. The beam is modeled using FEA, while the hand is modeled as a lumped parameter system. This approach is consistent with the methods described by Inman [10].

The combined system equations show how the beam and the Spring-Mass-Damper System interact:

$$\begin{bmatrix} m_{11} & 0 \\ 0 & m_{22} \end{bmatrix} \begin{bmatrix} X_m \\ X \end{bmatrix} + \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \begin{bmatrix} X_m \\ X \end{bmatrix} + \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} \begin{bmatrix} X \\ X_m \end{bmatrix} = \begin{bmatrix} f(t) \\ 0 \end{bmatrix}$$

Where:

- m_{11} represents the mass matrix of the finite element beam
- m_{22} represents the mass of the hand-arm system
- $c_{11}, c_{12}, c_{21}, c_{22}$ represent the damping coefficients of the combined system
- $k_{11}, k_{12}, k_{21}, k_{22}$ represent the stiffness coefficients of the combined system
- X_m represents the displacement of the beam
- X represents the displacement of the hand-arm system
- $f(t)$ represents the input force from the grinder

This representation allows us to analyze how vibrations are transmitted from the grinder through the handle to the operator's hand.

2. Frequency Domain Analysis of Combined System: We again transform the combined system to the frequency domain. This step is essential because it allows us to analyze the system's response across different frequencies and calculate the transmissibility, which is a key metric for evaluating vibration isolation performance.

The frequency domain equation for the combined system is:

$$\left(-\omega^2 \begin{bmatrix} m_{11} & 0 \\ 0 & m_{22} \end{bmatrix} + j\omega \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} + \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} \right) \begin{bmatrix} X \\ X_m \end{bmatrix} = \begin{bmatrix} f(t) \\ 0 \end{bmatrix}$$

For a harmonic input where we impose X_m is known to get X :

$$X_m(t) = X_m e^{j\omega t}$$

$$X(t) = X e^{j\omega t}$$

After solving the system of equations, we obtain:

$$X = (A)^{-1} \cdot B \cdot X_m$$

where

$$A = -\omega^2 \cdot m_{22} + \omega \cdot 1j c_{22} + k_{22}$$

$$B = -(\omega 1j c_{21} + k_{21})$$

Here, $X_m = 1$ To calculate Transmissibility, we find:

$$C = \text{inv}(A)$$

$$X = (C \cdot B) \cdot X_m$$

3. Transmissibility (R) Calculation: Calculating transmissibility is the ultimate goal of this analysis. Transmissibility (R) is the ratio of the output vibration to the input vibration. In our case, it's the ratio of the vibration experienced by the operator's hand (X_2) to the vibration at the grinder handle (X_1).

$$R = \frac{X_2}{X_1}$$

This ratio tells us how effectively the handle design reduces vibrations at different frequencies. A transmissibility less than 1 indicates vibration reduction, while greater than 1 indicates amplification. By calculating R across a range of frequencies, we can identify resonant frequencies where vibration amplification occurs. Evaluate the effectiveness of the handle design in reducing vibrations. And optimize the handle design by adjusting parameters to minimize transmissibility in the frequency range of concern. This analysis provides crucial insights for improving the ergonomics and safety of grinder handles, potentially reducing the risk of HAVs syndrome (HAVs) in operators.

4.2.2 Usage of LS-DYNA finite-element software

LS-DYNA is a advanced multi-physics simulation software used to model and analyze our SMDs system based on the standard [1]. This Finite Element Analysis software is particularly well-suited for dynamic problems involving complex interactions between multiple components.

In our LS-DYNA model, we utilized discrete elements to represent the springs and dampers. The masses were modeled using concentrated mass elements. This approach allows for accurate representation of the system’s dynamic behavior. We opted for an implicit analysis in LS-DYNA, which is particularly suitable for our FRF. Implicit analysis in LS-DYNA solves the equation of motion. The `CONTROL IMPLICIT EIGENVALUE` keyword in our LS-DYNA keyword file initiates an eigenvalue analysis. This analysis is crucial for understanding the natural frequencies and mode shapes of our system, which are fundamental to its dynamic response.

We used the `FREQUENCY DOMAIN FRF` keyword to perform a FRF. This analysis allows us to examine how our system responds to harmonic excitation across a range of frequencies, which is essential for comparing our results with the curves representing impedance in the standard (R1) and LS-DYNA generates output files containing mass, stiffness, and damping matrices. These matrices are crucial for our subsequent Python analysis and for validating our model against analytical solutions.

4.2.3 Usage of Python

Python served as our primary tool for post-processing LS-DYNA results and comparing them with analytical solutions. This approach combines the power of LS-DYNA’s FEA with Python’s flexibility in data manipulation and visualization.

We used a custom Python function (`lireMatriceRSA hb`) to read the mass, damping, and stiffness matrices from LS-DYNA output files. These matrices form the core of our system’s mathematical representation. The core of our Python analysis involves solving the equation of motion in the frequency domain for each frequency point. Our analysis covered a frequency range from 0 to 500 Hz, with a 1 Hz resolution. This range encompasses the frequencies of interest in HAV studies as specified in the standard [3].

4.3 Discussions

This section of study represents a comprehensive framework for analyzing HAV in grinder lateral handles. It integrates models of support hand-arm system based on the standard [1]. The analytical approach using a two-degree-of-freedom SMDs provides a fundamental understanding of the support hand-arm system dynamics. The frequency domain analysis enables the calculation of transmissibility. After that computational simulations use LS-DYNA to get mass, stiffness and damping matrices. By using all these three matrices, Python simulates the dynamic response of the system and plots frequency domain plots of the support hand-arm system amplitude. The results are consistent with the standard [1] model-B plots, validating that the model accurately reflects the vibration characteristics of the grinder lateral handle in real-world condition.

By using experimental stiffness data to create a robust approach for understanding and predicting vibration transmission. Integrating experimental data into the model and refining it with LS-DYNA and Python analysis has provided a comprehensive understanding of how different LFH designs affect vibration transmission. The accurate representation of the LFH dynamic behavior, combined with detailed FRF, has enabled us to identify effective strategies for reducing vibration and improving LFH. This approach not only validates our model against real-world conditions but also offers practical solutions for minimizing the risk of Hand-arm Vibration Syndrome (HAVs) in workers.

5 Conclusion and Future Works

This internship presents detailed study into effectiveness of Lateral Flexible Handles (LFH) in reducing Hand-arm Vibration (HAV) exposure for workers using electric grinder machines. By comparison with rigid lateral handle, our experiment demonstrates that the LFH reduce the vibration level to the operators hand up to 20-55%. However, the performance of LFH varies across LFH models and grinders models emphasizing the importance of designing LFH specifically tailored to individual grinder types to optimize vibration reduction.

Furthermore, from our experiment we can say that the LFH exhibit varying levels of vibration for different grinders. These differences of vibration reducing according grinder model are due to the dependence of vibration transmissibility on frequency. Some of LFH would be good at reducing low frequency vibrations, while others would be more effective at higher frequencies. This variability in performance across different frequency ranges highlights the complexity of vibration transmission in the grinder and underscores the need for the LFH design.

Our computational modeling, utilizing LS-DYNA Finite Element Analysis (FEA) and Python, provided important insights into dynamic behavior of the grinder-handle-hand system. This computational approach allows for rapid prototyping and testing of various LFH designs with grinders, potentially streamlining the development process for future LFH iterations.

The integration of the standard [1] into our model ensured that our analysis considered standardized representations of hand-arm system dynamics. This approach not only enhanced the models validity but also facilitates comparisons with other studies in the field.

Future research and development could go a number of directions. They include long-term studies to evaluate the effect on lowering the incidence of HAVs, exploration of smart materials for adaptive LFH designs, optimization of LFH designs for particular grinder models and operating frequencies, assessment of the functional impacts, and application of the concepts and methodologies to other kinds of hand-held power tools.

The substantial potential of LFH in reducing HAV exposure has been shown by this internship, in conclusion. This discipline has a strong foundation for future developments thanks to the combination of computational modeling and experimental validation. Understandings from this research will aid in the creation of new tools as occupational health issues continue to encourage development in LFH design.

References

Norme ISO 10068. Vibrations et chocs mécaniques — Impédance mécanique du système main-bras au point d'entrée, 2012.

Norme NF EN ISO 28927-1. Machines à moteur portatives — Méthodes d'essai pour l'évaluation de l'émission de vibrations — Partie 1 : meuleuses verticales et meuleuses d'angles, December 2019.

Norme NF EN ISO 5349-1. Mesurage et évaluation de l'exposition des individus aux vibrations transmises par la main, April 2002.

NF EN 60745-2-3. Hand-held motor-operated electric tools — Safety — Part 2-3: Particular requirements for grinders, polishers and disk-type sanders, 2011.

NF EN 60745-2-3. Outils électroportatifs à moteur — Partie 2-3 : Règles particulières pour les meuleuses, lustreuses et ponceuses du type à disque, April 2011.

Norme NF ISO 8662-4. Machine à moteur portatives — Mesurage des vibrations au niveau des poignées — Partie 4 : Meuleuses, July 1995.

Y. Aldien, D. Welcome, S. Rakheja, R. Dong, and P.-E. Boileau. Contact pressure distribution at hand–handle interface: role of hand forces and handle size. *International Journal of Industrial Ergonomics*, 35(3):267–286, 2005.

A.K. Chopra. *Dynamics of Structures: Theory and Applications to Earthquake Engineering*. Civil Engineering and Engineering Mechanics Series. Prentice Hall, 2012.

Ana Gomez, Jeff Volek, Martyn Rubin, Duncan French, Nicholas Ratamess, Matthew Sharman, and William Kraemer. Physiological and functional effects of acute low-frequency hand-arm vibration. *Journal of strength and conditioning research / National Strength & Conditioning Association*, 17:686–93, 11 2003.

D.J. Inman. *Engineering Vibration*. Pearson Education, Incorporated, 2014.

Thomas W. McDowell, Daniel E. Welcome, Christopher Warren, Xueyan S. Xu, and Ren G. Dong. The Effect of a Mechanical Arm System on Portable Grinder Vibration Emissions. *The Annals of Occupational Hygiene*, 60(3):371–386, 11 2015.

Ilmari Pyykkö, Martti Färkkilä, Jarmo Toivanen, Ok Korhonen, and Jonna Hyvärinen. Transmission of vibration in the hand arm system with special reference to changes in compression force and acceleration. *Scandinavian journal of work, environment & health*, 2:87–95, 07 1976.

D.D. Reynolds, R. Basel, D.E. Wasserman, and W. Taylor. A study of hand vibration on chipping and grinding operators, part i: Vibration acceleration levels measured on pneumatic tools used in chipping and grinding operations. *Journal of Sound and Vibration*, 95(4):479–497, 1984.

O.C. Zienkiewicz, R.L. Taylor, and J.Z. Zhu. Chapter 7 - elasticity: Two- and three-dimensional finite elements. In O.C. Zienkiewicz, R.L. Taylor, and J.Z. Zhu, editors, *The Finite Element Method: its Basis and Fundamentals (Seventh Edition)*, pages 211–255. Butterworth-Heinemann, Oxford, seventh edition edition, 2013.